

JEE MAIN REVISION MASTER: ALCOHOL, PHENOLS AND ETHERS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Ethers: Williamson Synthesis & Cleavage with HI	14 PYQs	Ethers
High	Phenols: Acidity & Named Reactions (Kolbe & Reimer-Tiemann)	13 PYQs	Phenols
Moderate	Alcohols: Preparation, Lucas Test & Dehydration	5 PYQs	Alcohols

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ethers: Williamson Synthesis & Cleavage with HI	$\text{Williamson: } \text{R-O}^- \text{Na}^+ + \text{R'-X} \xrightarrow{(1^\circ)} \text{R-O-R'}$ $\text{Cleavage of Anisole: } \text{C}_6\text{H}_5\text{OCH}_3 + \text{HI} \rightarrow \text{C}_6\text{H}_5\text{OH} + \text{CH}_3\text{I}$
Phenols: Acidity & Named Reactions (Kolbe & Reimer-Tiemann)	$\text{Kolbe: } \text{C}_6\text{H}_5\text{ONa} + \text{CO}_2 \xrightarrow{400^\circ\text{C}, 4-7\text{ atm}} \text{Salicylic Acid}$ $\text{Reimer-Tiemann: } \text{Phenol} + \text{CHCl}_3 / \text{NaOH} \rightarrow \text{Salicylaldehyde}$
Alcohols: Preparation, Lucas Test & Dehydration	$\text{Lucas: } 3^\circ \rightarrow \text{Immediate}, 2^\circ \rightarrow 5\text{ mins}, 1^\circ \rightarrow \text{No turbidity at room temp}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ethers: Williamson Synthesis & Cleavage with HI (High Yield — 14 PYQs)

Core Principle: Williamson Synthesis: Reaction of sodium alkoxide with 1° alkyl halide via S_N2 mechanism. Cleavage with concentrated HI: With unsymmetrical ethers containing 1° and 2° alkyl groups, the smaller alkyl group forms alkyl iodide via S_N2. If one alkyl group is 3° (e.g. tert-butyl methyl ether), the 3° group forms the iodide via S_N1 carbocation mechanism!

■ **Caution / JEE Pitfall:** In Williamson synthesis, using a 3° alkyl halide with sodium alkoxide yields an ALKENE via elimination, NOT an ether!

■ **Speed Trick:** Cleavage of anisole (extC_6 extH_5 extOCH_3) with HI always yields Phenol and Methyl Iodide because C_6H_5-O partial double bond cannot be broken.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The correct IUPAC name of is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Glycerol is a trihydric alcohol. It contains ____." [Ans: B]
- [2025 Shift] Q#1 (MCQ): "Ethyl alcohol is heated with concentrated sulphuric acid at 413 K. The major product..." [Ans: N/A]

3.2 Phenols: Acidity & Named Reactions (Kolbe & Reimer-Tiemann) (High Yield — 13 PYQs)

Core Principle: Phenol is acidic (pK_a ≈ 10) because phenoxide ion is resonance stabilized. Electron withdrawing groups (e.g. -NO₂) increase acidity; picric acid (2,4,6-trinitrophenol) is strongly acidic (pK_a = 0.38). Kolbe's reaction: Sodium phenoxide with CO₂ at 400 K and 4-7 atm gives salicylic acid (precursor to aspirin). Reimer-Tiemann reaction: Phenol with CHCl₃ + aq. NaOH generates dichlorocarbene electrophile (:CCl₂) to yield salicylaldehyde.

■ **Caution / JEE Pitfall:** The reactive electrophilic intermediate in Reimer-Tiemann reaction is DICHLOROCARBENE (:CCl₂), an electron-deficient neutral species with 6 valence electrons.

■ **Speed Trick:** Phenol reacts with bromine water to give a white precipitate of 2,4,6-tribromophenol; with neutral FeCl₃, it gives a characteristic violet coloration.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "Phenol can be distinguished from propanol by using the reagent..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Match the following with their pK_a values Acid pK_a (I) Phenol (a) 16 (II) p-Nitrophenol (b) 0.78 (III) Ethyl alcohol (c) 10 (I..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "A and B respectively are..." [Ans: N/A]

3.3 Alcohols: Preparation, Lucas Test & Dehydration (Moderate Yield — 5 PYQs)

Core Principle: Lucas reagent (anhydrous ZnCl_2 + conc. HCl) distinguishes alcohols: 3° alcohol gives turbidity immediately; 2° alcohol gives turbidity in 5 minutes; 1° alcohol gives no turbidity at room temperature.

Hydroboration-oxidation of alkenes gives anti-Markovnikov alcohol (without rearrangement). Acid-catalyzed dehydration of alcohols follows ease of carbocation: $3^\circ > 2^\circ > 1^\circ$.

■ **Caution / JEE Pitfall:** Oxidation of 1° alcohol with PCC (pyridinium chlorochromate) yields an aldehyde; oxidation with acidified KMnO_4 oxidizes it all the way to carboxylic acid.

■ **Speed Trick:** Methanol is poisonous because it is oxidized in the liver to methanal (formaldehyde) and then formic acid, which causes blindness and metabolic acidosis.

Representative Source PYQs:

• **[2026 Shift] Q#3 (MCQ):** " $R\text{-CH}_2\text{OH}$ is converted into $R\text{-CHO}$ by reacting with _____." [Ans: N/A]

• **[2026 Shift] Q#4 (MCQ):**

" $\text{C}_3\text{H}_6\text{O} \xrightarrow[\text{NaOH}]{\text{BH}_3 \cdot \text{THF}}$ $\text{C}_3\text{H}_7\text{O}_2\text{Na}$ " [Ans: N/A]

• **[2024 Shift] Q#2 (MCQ):** "When a tertiary alcohol ' $\text{C}_4\text{H}_{10}\text{O}$ ' reacts with $20\% \text{H}_3\text{PO}_4$ at 358 K , it gives a compound ' C_4H_8 ' as a major..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The correct IUPAC name of is

- (A) Tertiary butoxy ethane
- (B) 1,1-Dimethyl-1-ethoxyethane
- (C) 2-ethoxy-2-methyl propane
- (D) Ethoxy tertiary butane

Correct Answer: None

Solution Analysis: In an ether, the larger carbon part is chosen as the parent hydrocarbon, and the smaller part attached through oxygen is named as an alkoxy group. Here, one side of oxygen is an ethyl group, $-\text{C}_2\text{H}_5$. So this part becomes the alkoxy group: ethoxy, $-\text{O}-\text{C}_2\text{H}_5$. The other side is a tert-butyl group, $-\text{C}(\text{CH}_3)_3$. The longest continuous chain in this part has 3 carbon atoms, so the parent hydrocarbon is propane. In the propane chain, the middle carbon is carbon number 2. Both substituents are attached to this carbon. At carbon 2, there is one methyl group, $-\text{CH}_3$, and one etho

Benchmark Problem 2: KCET 2026 (Q#3)

$R-CH_2OH$ is converted into $R-CHO$ by reacting with _____.

- (A) Alkaline $KMnO_4$
- (B) $LiAlH_4$
- (C) Na/C_2H_5OH
- (D) PCC (Pyridinium Chlorochromate)

Correct Answer: None

Solution Analysis: $R-CH_2OH$ is a primary alcohol. To convert a primary alcohol into an aldehyde, we need a mild oxidising agent. PCC oxidises primary alcohols to aldehydes without further oxidising them to carboxylic acids.

$R-CH_2OH \xrightarrow{\text{Alkaline } KMnO_4} R-COOH$ Alkaline $KMnO_4$ is a strong oxidising agent, so it usually oxidises primary alcohols further to carboxylic acids: $R-CH_2OH \xrightarrow{Na/C_2H_5OH} R-CHO$ and $R-CH_2OH \xrightarrow{LiAlH_4} R-OH$

Benchmark Problem 3: KCET 2021 (Q#4)

A compound C_7H_8O is insoluble in $NaHCO_3$ solution but dissolve in $NaOH$ and give a characteristic colour with neutral $FeCl_3$ solution. When treated with bromine water compound 'A' forms the compound B with the formula $C_7H_5OBr_3$. 'A' is

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: It is given in the question that, compound A gives a characteristic colour with neutral $FeCl_3$ solution and soluble in $NaOH$ solution that means it contains a phenolic group. Since compound 'A' when treated with Br_2 forms compound 'B' with molecular formula $C_7H_5OBr_3$ (ppt.). Considering the given information the given compound is most likely to be m-cresol. i.e. Thus, the overall reactions are as follows.